

Research article

Integrated optimization for coordinated traffic signal and vehicle routing with parallel distributed solution

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ABSTRACT

Traffic signal control and vehicle routing are two promising strategies in traffic congestion management. However, conventional studies always treat these two strategies as separate modules, limiting the potential for system-wide efficiency gains. This study proposes an Integrated Traffic Signal and Route Guidance (ITSRG) optimization model in the Connected Vehicle (CV) environment. It simultaneously optimizes signal timing plans and individual vehicle routes. This study formulates the intrinsic coupling between signal plans and routing decisions as an explicit constraint, enabling the two variables to be jointly optimized within an integrated model. The proposed ITSRG model incorporates individualized route guidance for CVs, enabling full use of real-time vehicle information to achieve more accurate traffic control. Furthermore, to efficiently solve the complex integrated model, a parallel distributed solution algorithm based on the Alternating Direction Method of Multipliers (ADMM) algorithm is proposed. It decomposes the integrated problem into multiple subproblems, including the intersection signal control subproblem and the individual CV route selection subproblem, enabling scalable computation and real-time applicability in large networks. To validate the model's effectiveness, experiments were conducted on a real-world road network in Qinzhou City, China. The results demonstrate that the proposed ITSRG model significantly outperforms benchmark methods in reducing average vehicle travel time. Furthermore, sensitivity analyses are conducted to evaluate the model's performance under varying traffic demand levels, compliance rates, and parameters. Moreover, the effectiveness and computational efficiency of the proposed parallel distributed algorithms are validated through numerical experiments.

1. Introduction

Effective traffic management plays a vital role in mitigating increasingly severe urban congestion (Jialiang et al., 2023; Yang et al., 2026). Traditionally, traffic management research has focused on two main directions: traffic signal control and vehicle routing (Yang et al., 2025). Traffic signal control adjusts signal timing plans at the intersection level to reduce vehicle delay and queue length, while vehicle routing aims to select the optimal route with minimum travel cost. In real-world implementation, as shown in Fig. 1, traffic signal control and vehicle routing are highly coupled. Traffic signal control directly affects vehicle delay (as shown in Fig. 1(a)), which determines link travel times and thereby influences vehicle routing decisions. On the other hand, vehicle route selection reshapes the traffic demand distribution (as shown in Fig. 1(b)), which in turn determines the optimal signal plan (Chen & Qiu, 2022).

Limited by traffic state sensing and communication technologies, traditional traffic management research often treats signal control and vehicle routing as separate modules. Early vehicle routing problems focused on aggregated traffic flow assignment (Ghali & Smith, 1995). However, the importance of signal control in influencing route choices has been overlooked. Traditional traffic signal control methods mainly focus on optimizing signal timing plans based on historical traffic demand patterns to improve local intersection performance. These methods typically assume fixed and aggregated route behavior and fail to capture the dynamic coupling relationship between traffic signals and vehicle route choices within the network. Therefore, coordinating vehicle routing and traffic signal control is essential to improve traffic management efficiency.

The coordinated signal control and vehicle routing problems have been widely investigated in existing studies. The coordination relies

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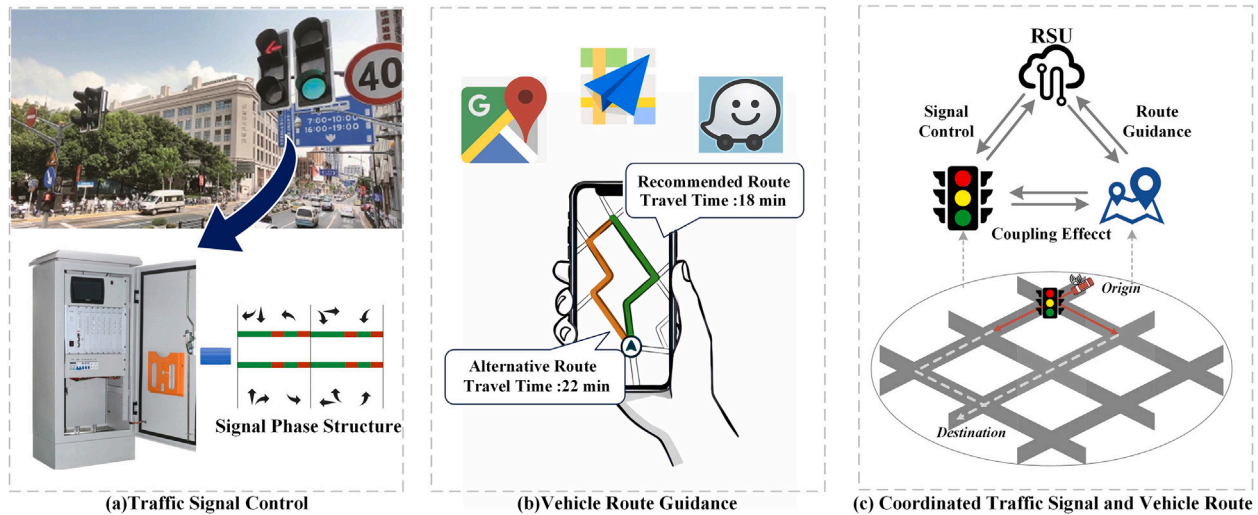


Fig. 1. The engineering background of traffic management.

on RoadSide Units (RSUs) to facilitate information exchange and decision coordination among intersections and vehicles (Zhu et al., 2025), as shown in Fig. 1(c). The coordinated problem is typically formulated as two sequential optimization models. The first-stage (or upper level) problem focuses on traffic flow assignment, while the second-stage (or lower level) optimizes network signal plans based on the determined vehicle routes (Lin et al., 2024; Sun et al., 2006). However, the two-stage framework still optimizes signal timing and route guidance independently. Signal plans optimized in the second stage may invalidate the user equilibrium or system optimal condition assumed in the first stage. As a result, it fails to achieve effective coordination. Moreover, in these coordinated optimization methods, the decision variable of vehicle routing is always the aggregated turning rate in flow assignment (Hajbabaie & Benekohal, 2015). It only provides a macroscopic representation of traffic flow and lacks refined guidance for individual vehicles. The optimized aggregated turning rates fail to guide actual vehicle behavior, leading to noticeable discrepancies between the optimization results and real traffic dynamics (Pengfei et al., 2015). The main challenge of individual vehicle route guidance lies in its high computational complexity. Optimizing vehicle-level routes requires simultaneous consideration of numerous interacting vehicles and dynamic signal plans, which exponentially increases the decision space and renders centralized computation impractical for large-scale networks. A comprehensive review of existing studies on traffic signal control and vehicle routing is presented in Section 2.

Despite significant advances in traffic management studies, several challenges remain unresolved. First, most existing studies treat signal control and route planning as two separate tasks, which ignore the reciprocal influence between vehicle routes and signal operations, leading to suboptimal results. Second, most existing studies usually focus on aggregate flow assignment in the optimization problem. It cannot represent the discrete and heterogeneous nature of vehicle decisions. Advancements in Connected Vehicle (CV) technologies and mobile navigation systems (Feng et al., 2018) allow for precise, real-time routing decisions at the vehicle level. Third, the huge computational burden of the integrated signal-route optimization is a concern. As the decision space grows with network size and vehicle number, the integrated model becomes computationally intractable for real-time optimization, limiting its applicability in large urban networks.

To address these research gaps, this study proposes an Integrated Traffic Signal and Route Guidance (ITSRG) model. The ITSRG model simultaneously optimizes intersection signal plans and individual vehicle routes under the CV environment. To efficiently solve this complex problem, a parallel distributed algorithm based on Alternating Direction

Method of Multipliers (ADMM) is developed. The original ITSRG model is reformulated as a consensus optimization problem and decomposed into signal control, individual CV routing, and RSU-based coupling subproblems. Specifically, a Jacobi-style update scheme is adopted for these subproblems, allowing them to be solved simultaneously without sequential dependency. Global coordination is achieved through consensus and dual updates, enabling scalable and efficient computation.

The contributions of this paper are listed below:

- This study proposes an integrated optimization model for coordinated signal control and vehicle routing at the network level. The model captures the coupling interactions between the traffic signals and vehicle routing in an explicit coupling constraint. It ensures that traffic signals and route guidance are optimized simultaneously, improving their coordination.
- This study incorporates individualized vehicle route guidance into the integrated model, rather than only optimizing aggregated traffic flow assignment. Leveraging the distributed computational resources of CVs, it provides route guidance at the individual-vehicle level and allows the model to make full use of CV information for more accurate traffic control.
- This study develops a problem-tailored parallel solution algorithm for the integrated signal-route coordination problem. We propose a modified ADMM framework with a Jacobi-type parallel update scheme. To further tackle the computational difficulty, we develop a consensus update design based on the continuous variable. This algorithm decomposes the ITSRG problem into signal control and vehicle routing subproblems that can be solved in parallel.

This study is structured as follows: Section 2 discusses related work, while Section 3 focuses on the coordination of signal control and vehicle routing. Section 4 presents the proposed ITSRG model formulation. Section 5 introduces the parallel solution algorithms for the ITSRG model. Section 6 reports simulation experiments and results discussion. Section 7 concludes this study and proposes future research directions.

2. Related work

In this section, we first review literature on traffic signal control and vehicle route planning, and then introduce the study on coordinated signal control and vehicle routing.

2.1. Traffic signal control

Early studies on traffic signal control focused on fixed-time strategies at the isolated intersection (Allsop, 1972). It was soon recognized

that optimizing signals in isolation is insufficient to achieve efficient network-wide performance. Network-level signal coordination has long been recognized as an important research direction (Little, 1966). Seminal studies formulated coordinated signal timing for multiple intersections using mathematical programming (Gartner et al., 1991). However, as network size increases, these centralized models face significant challenges in scalability and real-time implementation (Aboudolas et al., 2009; Mirchandani & Head, 2001).

With the rapid progress of parallel and distributed computing (Adeli & Kumar, 2020; Hung & Adeli, 1993), recent studies have increasingly shifted toward decentralized signal-control frameworks. These approaches decompose the network-level optimization problem into multiple intersection-level subproblems. Each local controller optimizes its own signal plan while coordinating with neighboring intersections through limited information exchange (Luo et al., 2025; Varaiya, 2013). For example, Noaeen et al. (2021) developed a decentralized signal control method based on the kinematic wave theory. More recently, Zhu, Chengyuan et al. (2025) proposed an asynchronous decentralized coordination framework, in which each intersection updates its signal decisions using the most recent information received from neighboring controllers. Despite these advancements, conventional models typically assume static route choices or historical turning ratios. This formulation neglects the dynamic coupling between signal timing and vehicle routing decisions.

2.2. Vehicle route planning

Early studies on vehicle route planning focused mainly on shortest path search algorithms, such as Dijkstra's and A^* , which efficiently identify the minimum-cost path in static networks (Bertsimas, 1992; Psaraftis, 1995). For example, Bayili and Polat (2011) proposed an improved A^* algorithm that improves computational efficiency and precision. Smolic-Rocak et al. (2009) combined Dijkstra's algorithm with the dynamic window approach. However, these shortest path algorithms often assume a fixed travel time for each link and neglect interactions among multiple vehicles' route choices, which limits their applicability in congested urban networks.

Therefore, related studies shifted toward traffic assignment models that incorporate aggregated flow dynamics. These models account for route choice interactions between vehicles by using the aggregated model. Many researchers studied Dynamic Traffic Assignment (DTA)-based models to assign traffic flow across the network, where Dynamic User Equilibrium (DUE) methods capture individual route choice rationality, and Dynamic System Optimum (DSO) methods focus on overall network efficiency (Satsukawa et al., 2022). However, these models typically rely on aggregate flow representations and fail to integrate real-time traffic conditions and heterogeneous driver choices into route planning.

The CV technologies enable real-time, individualized route guidance. Kim et al. (2016) proposed a dynamic CV routing method integrating time-varying probabilistic travel times under congested conditions. Guo et al. (2018) proposed a route planning algorithm based on real-time travel time estimation. However, most prior studies fail to account for how optimized vehicle routes subsequently affect signal control (Qadri et al., 2020). Overlooking this dependency leads to inaccurate route cost estimates and poor coordination between routing and signal systems.

2.3. Coordination of signal control and vehicle routing

The coordination between signal control and vehicle routing has been widely studied. Most existing studies formulate this coordination as two separate optimization problems. The first determines vehicle routes or turning flows, typically using DUE or DSO models. The second optimizes signal control, adopting fixed time, actuated, or optimization-based strategies (Peng & Wang, 2023). For example, Sun et al. (2006) proposed a bi-level model, where the upper level models signal control and the lower level represents user travel behavior. Hao et al. (2018)

developed an optimal signal control model under DUE conditions, integrating signal control and traffic flow assignment through a genetic algorithm. Chen and Qiu (2022) proposed a joint framework for route guidance and signal control. Vehicle routing follows the UE or SO principle. Signal timings are then adjusted adaptively using traffic volumes from adjacent intersections.

In recent years, many researchers have been dedicated to establishing integrated models that combine signal control and vehicle routing to ensure better coordination. Hajbabaie and Benekohal (2015) proposed a model for optimizing network-wide signal timing and traffic assignment, incorporating link flow allocation through a genetic algorithm. Safadi and Haddad (2021) proposed a continuous-time model that integrates traffic assignment and signal control for a simple traffic network. However, most existing studies optimize aggregated flow assignment or turning rates. Such formulations cannot generate route guidance for individual vehicles. As a result, their practical applicability is limited by the mismatch between the optimization results and heterogeneous driver behavior.

In the CV environment, Huajun Chai et al. (2017) proposed a joint optimization of individual route choice and signal control by continuously updating all link travel times in the network. Nevertheless, the computational burden remains considerable, as the route costs must be repeatedly evaluated for each traveler at every time interval. Chow et al. (2020) proposed an adaptive decentralized traffic control coupled with a user-optimal routing process to generate individualized routes. However, the iterative recalculation of optimal paths results in a substantial computational burden. Recently, Lin et al. (2024) proposed a two-stage optimization model for individual CV route and signal control. Each intersection determines its local signal plan and the routes of CVs passing through it. Due to the model complexity and computational burden, some studies restrict individualized routing and signal control to scenarios with a limited subset of vehicles, notably emergency vehicles or other high-priority vehicles (Humagain et al., 2020; Nguyen et al., 2021).

Recent advances in Reinforcement Learning (RL) offer a promising method for the coupling between signal control and vehicle routing. For example, Shi et al. (2023) proposed a model-free deep RL approach that accounts for the influence of signal control in adaptive route guidance. Chang et al. (2025) developed a hierarchical RL framework that independently optimizes vehicle routing and intersection signal control. However, these methods typically treat strict coupling constraints as soft penalty terms. As a result, their performance is often sensitive to training configurations and may become unstable in highly dynamic traffic environments.

A summary of representative literature on coordination of signal control and vehicle routing methods is provided in Table 1.

3. Preliminaries

3.1. Network representation and vehicle definition

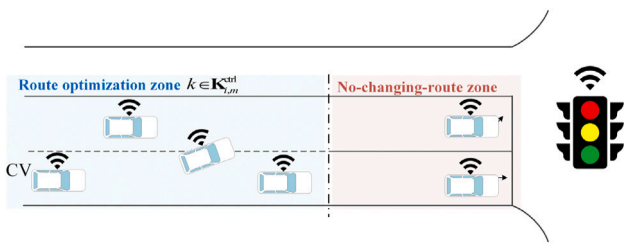
Consider an urban road network modeled as a directed graph $G = (\mathcal{N}, \mathcal{L})$, where $\mathcal{N}$ and $\mathcal{L}$ denote the sets of signalized intersections (nodes) and links, respectively. For each intersection $i \in \mathcal{N}$, M_i denotes the set of feasible movements. In this study, we adopt the decentralized signal control strategy, where each signalized intersection is managed by a signal controller that dynamically adjusts the local signal plan. The coordination between adjacent intersections is achieved through the exchange of predicted arrival flows derived from individualized CV route guidance, enabling each intersection to predict upstream traffic and optimize its local signal plans accordingly (Islam & Hajbabaie, 2017; Zhu, Chengyuan et al., 2025).

It is assumed that all vehicles are CVs, each equipped with computational resources (i.e., CV controllers) such as On-Board Units (OBUs) and advanced navigation software, enabling real-time route adjustment. In addition, RSUs are assumed to be deployed to handle information exchange and coordination between traffic signal controllers and CV

Table 1

Summary of representative studies on traffic signal control, vehicle route planning, and their coordination optimization.

Category	Research Innovation and Limitation	Control Variable	Solution Algorithm
Traffic Signal Control			
Putha et al. (2012)	A decentralized signal control model for queue dissipation in oversaturated networks, with high computational effort and offline-oriented optimization.	Signal plan	Ant Colony Algorithm
Wada et al. (2018)	A centralized signal control model optimizing network-wide signals under deterministic and stochastic demand, assuming exogenous traffic routing.	Signal plan	Approximation-based algorithm
Zhu, Chengyuan et al. (2025)	An asynchronous decentralized signal control model with rolling-horizon MILP, involving nontrivial parameter calibration.	Signal plan	Decentralized solution
Vehicle Route Planning			
Molina et al. (2014)	A multi-objective routing model with heterogeneous fleets and emissions, without traffic dynamics or signals.	Individual vehicle route	Weighted algorithm
Guo et al. (2018)	A dynamic route planning method based on real-time traffic information sharing, with no consideration of the signal.	Individual vehicle route	Analytical solution
Satsukawa et al. (2022)	A dynamic traffic assignment problem formulated at the link-flow levels, lacking individual vehicle-level routing decisions.	Link flow assignment	Evolutionary algorithm
Coordination of Signal Control and Vehicle Routing			
Sun et al. (2006)	A bi-level optimization model that links signal control with dynamic traffic assignment. Its method solves the two problems separately.	Signal plan; link flow assignment	Genetic Algorithm
Hajbabaie and Benekohal (2015)	An integrated model jointly optimizing signal timing and traffic assignment, relying on centralized heuristic optimization.	Signal plan; link flow assignment	Genetic algorithm
Zaidi et al. (2016)	A two-stage framework integrating back-pressure signal control and adaptive vehicle routing. Its method solves the two problems separately.	Signal plan; individual vehicle route	Decentralized solution
Safadi and Haddad (2021)	An integrated model coupling signal control and traffic assignment, primarily applicable to simplified network settings.	Signal plan; traffic assignment	Analytical solution
Chen and Qiu (2022)	A two-stage framework for joint signal control and dynamic route guidance, where the sequential logic and slow solution process limit effectiveness.	Signal plan; individual vehicle route	Analytical solution
Lin et al. (2024)	A two-stage framework coordinating signal control and individual vehicle routing, involving sequential decision-making across stages.	Signal plan; individual vehicle route	Decentralized solution
ITSRG model in this study	An integrated model enabling real-time coordination of signal control and individual vehicle routing through a parallel distributed algorithm.	Signal plan; individual vehicle route	Parallel distributed algorithm

**Fig. 2.** Illustration of the route optimization zone and the no-changing-route zone.

routing decisions. For modeling tractability, in this study, we assume an ideal communication environment where the information exchange among the signal controller, CVs, and RSUs is timely and accurate. Unlike conventional methods that optimize aggregated flow assignment, our method directly determines the route guidance for each CV. To enable real-time optimization, our model determines each CV's direction at the current intersection (through, left-turn, and right-turn) in the optimization horizon p .

Each link is divided into two zones (as shown in Fig. 2): the route optimization zone and the no-changing-route zone. The no-changing-route zone refers to a downstream segment of the link with a fixed length L^{NC} , measured upstream from the stop line, where route changes are operationally infeasible due to safety considerations and lane-use restrictions

(Yuqi Shi et al., 2025). For a vehicle k approaching intersection i , let $d_{i,k}$ denote its distance to the stop line. If $d_{i,k} \leq L^{\text{NC}}$, the vehicle is considered to be within the no-changing-route zone. In this zone, vehicles are assumed to have committed to their current route guidance (i.e., selected turning direction), and their routing decisions are frozen and excluded from further optimization at the current intersection. The set of all vehicles associated with movement m at intersection i is denoted by $\mathbf{K}_{i,m}$. In contrast, the subset of vehicles eligible for routing optimization is denoted as $\mathbf{K}_{i,m}^{\text{ctrl}} \subseteq \mathbf{K}_{i,m}$. The choice of L^{NC} directly affects the size of $\mathbf{K}_{i,m}^{\text{ctrl}}$ and the trade-off between routing flexibility and maneuver feasibility. In practice, the setting of L^{NC} is typically constrained by roadway design standards, driver perception–reaction time, vehicle travel speed, and feasible maneuver preparation distances (Yuqi Shi et al., 2025). Its impact on our model performance is systematically examined through a sensitivity analysis in Section 6.4.3.

3.2. Rolling horizon optimization scheme

To accommodate dynamic traffic conditions and ensure real-time adaptability, the proposed ITSRG model employs a rolling horizon optimization scheme (Chunhui et al., 2020). The scheme decomposes the entire period into multiple shorter horizons. Let p denote the index of an optimization horizon. At the start of each horizon p , the optimization problem is re-solved using the most recent traffic state information collected from CV controllers and signal controllers. The updated solution provides the optimal signal timing plan and route guidance for the

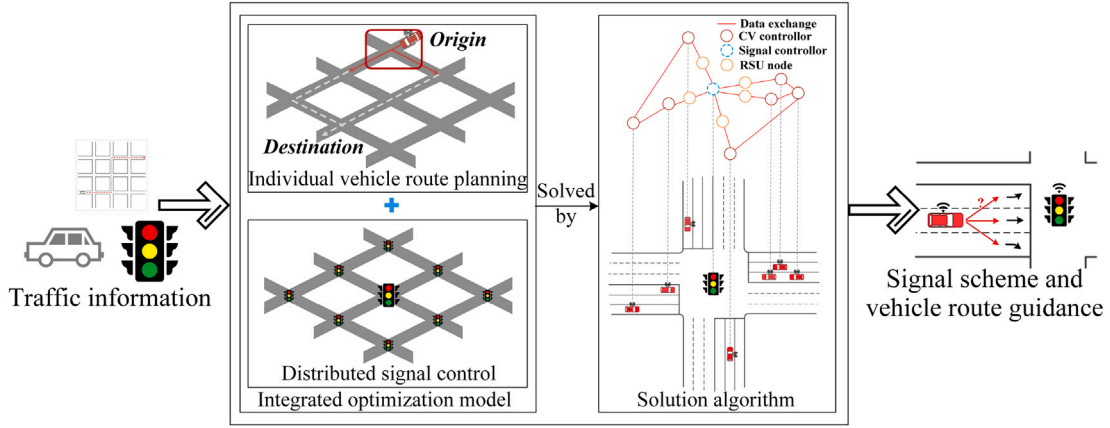


Fig. 3. The overall framework of the proposed methodology.

optimization horizon p . Future signal timing plans and route guidance are continuously adjusted as new data become available. This scheme maintains responsiveness to evolving traffic conditions while avoiding excessive computational burden (Chunhui et al., 2018). It also mitigates the impact of prediction errors and improves control robustness and stability.

4. The ITSRG model formulation

As illustrated in Fig. 3, the ITSRG model uses real-time traffic information, including vehicle locations, destinations, and speeds, as inputs. The model described in Section 4 formulates the integrated optimization of intersection signal timing and vehicle route guidance. To meet real-time operational requirements, a parallel distributed algorithm decomposes the integrated problem into multiple subproblems to solve, as detailed in Section 5. The ITSRG model outputs the signal timing plans for each intersection and the individual route choices for all CVs in the network. This section outlines the formulation of the ITSRG model. The model is developed under a decentralized signal control structure, where each controller optimizes only the local signal timing and the route choices of CVs within its intersection. The notation of the proposed ITSRG model is provided in Appendix A.

4.1. Decision variable

The decision variable of each signal controller is $u_s = [s_{i,m}^p, e_{i,m}^p]_{m \in \mathbf{M}_i}$, where $s_{i,m}^p$ and $e_{i,m}^p$ denote the start and end times of green duration of movement $m \in \mathbf{M}_i$ in the p th optimization horizon. $\mathbf{M}_i$ is the set of movements of intersection i . The signal control problem is modeled in the decentralized approach, where each intersection controller only optimizes the local signal plan (Zhu, Chengyuan et al., 2025).

The decision variable of CV controller k is $u_k = [\sigma_{i,m,k,d}^p, \zeta_{i,m,k}^p]$, where $\sigma_{i,m,k,d}^p$ is a binary variable to represent vehicle k on movement m at intersection i whether choose the direction $d \in D$ in the p th optimization horizon. The set $D = \{\text{left, through, right}\}$ is the set of directions. When the vehicle k selects direction d , then $\sigma_{i,m,k,d}^p = 1$; otherwise, $\sigma_{i,m,k,d}^p = 0$. The route strategy of each CV k is selecting one direction $d \in D$ at each intersection. $\zeta_{i,m,k}^p$ represents the travel cost of vehicle k on movement m at intersection i , which is determined by route selection variable $\sigma_{i,m,k,d}^p$.

4.2. Objective function

The objective of the ITSRG model is to reduce both intersection i 's delay (J_i) and each vehicle k 's travel cost (J_k).

4.2.1. Signal control objective

The objective is to minimize the total vehicle delay D_i^p at the intersection. To mathematically formulate this, the intersection-level delay

is expressed as the aggregation of individual vehicle delays within the optimization horizon p . Following the definition in traffic flow theory, the delay d_k^p for an individual vehicle k is defined as the difference between its actual departure time τ_k^p and its virtual arrival time t_k^p at the stop line:

$$D_i^p = \sum_{m \in \mathbf{M}_i} \sum_{k \in \mathbf{K}_{i,m}} d_k^p \quad (1)$$

$$d_k^p = \tau_k^p - t_k^p, \quad \forall m \in \mathbf{M}_i, k \in \mathbf{K}_{i,m} \quad (2)$$

$$\tau_k^p = s_{i,m}^p + p_k h_m + t_{\text{loss}}, \quad \forall m \in \mathbf{M}_i, k \in \mathbf{K}_{i,m} \quad (3)$$

where t_k^p is a constant determined by the vehicle's arrival profile. For a vehicle k in the movement m , the actual departure time τ_k^p depends on the green start time $s_{i,m}^p$. It is determined by the queue position p_k of each queued vehicle. The calculation further incorporates two parameters: the average saturated headway h_m , and the start-up lost time t_{loss} , which jointly characterize the discharge process at the intersection.

As formulated in Tan and Yang (2024), the total delay D_i^p can be simplified to $\hat{D}_i^p$ in Eq. (4). This allows the original optimization problem to be reformulated in a reduced form that explicitly depends on signal timing decisions only. This substitution decomposes the total delay D_i^p into two parts: a component that depends on signal timing decisions and a constant component. This is a widely used delay approximation in traffic signal optimization problems (Tan et al., 2025; Yao et al., 2019).

$$\hat{D}_i^p = \sum_{m \in \mathbf{M}_i} \sum_{k=1}^{q_{i,m}^{0,p}} (s_{i,m}^p + p_k h_m + t_{\text{loss}} - t_k^p) = \sum_{m \in \mathbf{M}_i} q_{i,m}^{0,p} s_{i,m}^p + C_{\text{const}} \quad (4)$$

where C_{const} represents delay components determined by historical arrivals and saturation flow, which are independent of the current optimization decision. Therefore, minimizing total delay is mathematically equivalent to minimizing the variable term $q_{i,m}^{0,p} s_{i,m}^p$.

To prevent queue spillover, we incorporate a residual queue penalty. The general objective function of the ITSRG model is written as:

$$J_i^p = \sum_{m \in \mathbf{M}_i} q_{i,m}^{0,p} s_{i,m}^p + \pi \lambda_{i,m}^p \quad (5)$$

where the residual queue $\lambda_{i,m}^p$ represents the number of vehicles remaining undischarged at the end of the optimization horizon, as detailed in Eq. (12). π is a weighting coefficient converting the residual queue $\lambda_{i,m}^p$ into equivalent delay units.

4.2.2. Route guidance objective

The objective function in Eq. (6) aims to minimize the travel cost of each vehicle $k \in \mathbf{K}_{i,m}^{\text{ctrl}}$. The formulation of J_k is given as follows:

$$J_k^p = \zeta_{i,m,k}^p, \quad \forall m \in \mathbf{M}_i, k \in \mathbf{K}_{i,m}^{\text{ctrl}} \quad (6)$$

where $\zeta_{i,m,k}^p$ denotes the travel cost incurred by vehicle k during the p th optimization horizon.

4.3. Model constraints

4.3.1. Signal plan constraints

This section details the formulation of the signal plan constraints. Each intersection i optimizes the local signal plan based solely on the outflow information from neighboring intersections.

Traditional fixed-phase structures act as rigid constraints, reducing the solution space and limiting vehicle route guidance flexibility (Chunhui et al., 2018). In contrast, a flexible phase structure relaxes these rigid constraints, dynamically adjusts cycle lengths and phase sequences based on real-time vehicle routing information, and enables more efficient utilization of intersection space-time resources (Feng et al., 2018). Therefore, we adopt a flexible signal scheme without the fixed phase structure and sequence as shown in Eqs. (7) and (8):

$$s_{i,n}^p \geq e_{i,m}^p + \psi - M(2 - \gamma_{m,n} - \phi_{m,n}), \quad \forall m, n \in \mathbf{M}_i \quad (7)$$

$$e_{i,n}^p \leq s_{i,m}^p - \psi + M(1 - \gamma_{m,n} + \phi_{m,n}), \quad \forall m, n \in \mathbf{M}_i \quad (8)$$

where $\gamma_{m,n}$ is a binary parameter representing the incompatibility between movements m and n . If movements m and n are incompatible, then $\gamma_{m,n} = 1$; otherwise, $\gamma_{m,n} = 0$. Similarly, let $\phi_{m,n}$ be a binary variable indicating the sequence of movements. If the phase of movement n occurs after movement m , then $\phi_{m,n} = 1$; otherwise, $\phi_{m,n} = 0$. The parameter ψ denotes the safety time gap between conflicting green phases.

For each movement m at intersection i , the green duration must remain within the interval delimited by the lower and upper bounds. The corresponding constraints are formulated as follows:

$$g_{\max} \geq e_{i,m}^p - s_{i,m}^p \geq g_{\min}, \quad \forall m \in \mathbf{M}_i \quad (9)$$

where $g_{\min}$ and $g_{\max}$ denote the minimum and maximum allowable green durations. Within the optimization horizon p , each movement m is permitted to receive a green phase no more than once, and some movements may not be assigned any green duration.

This study adopts a rolling horizon scheme, where the signal control plan is re-optimized at each optimization horizon p . A key challenge is that frequent re-optimizations can introduce discontinuities in signal plans between consecutive optimization horizons (Chengyuan et al., 2023). Such abrupt changes can disrupt vehicle progression and compromise traffic stability. To mitigate these adverse effects, the ITSRG model provides signal plan continuity constraints. Specifically, the signal phase that is active at the end of the previous horizon $p-1$, is treated as a fixed initial state for the current optimization horizon p . The corresponding constraint is shown in Eq. (10):

$$e_{i,m}^p = e_{i,m}^{p-1} \left| \left(s_{i,m}^{p-1} \leq t_p \text{ and } e_{i,m}^{p-1} > t_p \right) \right|, \quad \forall m \in \mathbf{M}_i \quad (10)$$

Eq. (11) enforces a safety green interval ψ between conflicting phases across consecutive optimization horizons to ensure safe transitions. If phase n from the previous optimization plan extends beyond the current horizon boundary t_p , the start of conflicting phase m must be delayed until $(e_{i,n}^{p-1} + \psi)$ to preserve the required safety gap.

$$s_{i,m}^p \geq e_{i,n}^{p-1} + \psi - M(1 - \gamma_{n,m}) \left| \left(e_{i,n}^{p-1} > t_p \right) \right|, \quad \forall m, n \in \mathbf{M}_i, m \neq n \quad (11)$$

The traffic dynamics at signalized intersections are commonly represented using a point-queue model (Jin, 2015), which characterizes the evolution of queue lengths over discrete time steps. The evolution of queue length $\lambda_{i,m}^p$ is formulated as follows:

$$\lambda_{i,m}^p = \max \left\{ \lambda_{i,m}^{p-1} + q_{i,m}^{\text{in},p} - q_{i,m}^{\text{out},p}, 0 \right\}, \quad \forall m \in \mathbf{M}_i \quad (12)$$

where $\lambda_{i,m}^p$ represents the residual queue vehicles for movement m at intersection i in the optimization horizon p . $q_{i,m}^{0,p}$ denotes the initial number

of queued vehicles for movement m at intersection i at the beginning of the p th optimization horizon; $q_{i,m}^{\text{in},p}$ denotes the number of vehicles arriving at the stop line for movement m at intersection i during the optimization horizon p . $q_{i,m}^{\text{out},p}$ denotes the number of vehicles that depart the stop-line for movement m at intersection i during the optimization horizon p .

The inflow $q_{i,m}^{\text{in},p}$ to movement m at intersection i during horizon p consists of vehicles already present at the intersection and those arriving from upstream movements. As shown in Eq. (13), the first term represents vehicles that are already located in intersection i and assigned to movement m during optimization horizon p , which is determined by the $\sigma_{i,m,k,d}^p$. The second term represents the inflow arriving from the upstream intersections. This component originates from the set of upstream movements U_m . This inflow is determined by a dynamic turning rate, $\phi_{i,m}^p$. Eq. (13) establishes the coupling relationship between signal control and vehicle routing: the route decision variable $\sigma_{i,m,k,d}^p$ determines the inflow of each movement, and the resulting inflow directly affects the optimal signal control plan. The relevant formulations are as follows:

$$q_{i,m}^{\text{in},p} = \sum_{k \in \mathbf{K}_{i,m}} \sum_{d \in D} \sigma_{i,m,k,d}^p + \phi_{i,m}^p \sum_{n \in U_m} q_{i,n}^{\text{out},p-1}, \quad \forall m \in \mathbf{M}_i \quad (13)$$

where the inflow $q_{i,m}^{\text{in},p}$ to movement m at intersection i during horizon p consists of two components. The first component represents vehicles already present at intersection i that are assigned to movement m within the current optimization horizon. This assignment is described by the routing decision variable $\sigma_{i,m,k,d}^p$. The second component represents vehicles arriving from upstream intersections. It is generated by upstream movements $n \in U_m$ and scaled by the dynamic turning rate $\phi_{i,m}^p$, which specifies the proportion of upstream outflow directed toward movement.

The $\phi_{i,m}^p$ is obtained from the vehicle routing decisions at the upstream intersection j during the previous optimization horizon $p-1$ (Lin et al., 2023), as defined in Eq. (14).

$$\phi_{i,m}^p = \frac{\sum_{n \in U_m} \sum_{k \in \mathbf{K}_{j,n}} \sum_{d \in D} \sigma_{j,n,k,d}^{p-1}}{\sum_{m' \in \mathbf{M}_j} \sum_{k \in \mathbf{K}_{j,m'}} \sum_{d \in D} \sigma_{j,m',k,d}^{p-1}}, \quad \forall m \in \mathbf{M}_i \quad (14)$$

where $\phi_{i,m}^p$ denotes the fraction of vehicles departing from upstream movement n that select downstream movement m during horizon p . The $\sigma_{j,n,k,d}^{p-1}$ associated with vehicles departing from upstream movements $n \in U_m$ and directed toward movement m at intersection i are aggregated. This aggregated value is normalized by the total outflow from the upstream intersection j , thereby yielding the proportion of upstream traffic contributing to the demand for movement m at intersection i .

The outflow $q_{i,m}^{\text{out},p}$ is defined as the vehicle departing volume for movement m at intersection i during the p th optimization horizon. The discharge flow is determined by both traffic demand and traffic supply. Specifically, traffic demand refers to the number of vehicles seeking to traverse the intersection, while traffic supply represents the number of vehicles that can be discharged under the optimized signal plan, as expressed in Eq. (15).

$$q_{i,m}^{\text{out},p} = \min \left\{ q_{i,m}^{0,p} + q_{i,m}^{\text{in},p}, S_m (e_{i,m}^p - s_{i,m}^p - t_{\text{loss}}) \right\}, \quad \forall m \in \mathbf{M}_i \quad (15)$$

where S_m denotes the saturation flow rate, treated as a fixed parameter in the model. In this study, S_m is set to 0.5 veh/s/lane to provide a consistent baseline; However, recognizing that saturation flow varies by geometry and traffic composition, our model allows S_m to be calibrated as a specific parameter for the lane-group in real-world deployments using field data.

4.3.2. Vehicle routing constraints

To ensure that each vehicle selects only one feasible route, Eq. (16) imposes a route exclusivity constraint. It restricts the summation of all

route selection variables associated with vehicle k to unity.

$$\sum_{d \in D} \sigma_{i,m,k,d}^p = 1, \quad \forall m \in \mathbf{M}_i, k \in \mathbf{K}_{i,m}^{\text{ctrl}} \quad (16)$$

Eq. (17) defines the travel cost variable $\zeta_{i,m,k}^p$ as a continuous variable representing the realized travel cost incurred by vehicle k at intersection i . The value of $\zeta_{i,m,k}^p$ is endogenously determined by the selected route through the binary route selection variables $\sigma_{i,m,k,d}^p$. Specifically, $\zeta_{i,m,k}^p$ is expressed as a weighted sum of the candidate route costs, where the weights are given by the corresponding route selection variables.

$$\zeta_{i,m,k}^p = \sum_{d \in D} \sigma_{i,m,k,d}^p T_{i,m,k,d}^p, \quad \forall m \in \mathbf{M}_i, k \in \mathbf{K}_{i,m}^{\text{ctrl}} \quad (17)$$

where $T_{i,m,k,d}^p$ is the travel cost if vehicle k selects direction d . It is determined by the travel cost of the shortest route when the vehicle k selects direction d .

Traditionally, the travel cost $T_{i,m,k,d}^p$ is defined as the cumulative travel time of successive downstream links. Existing literature offers diverse methodologies for link travel time estimation, ranging from analytical functional forms to machine learning models (Abdollahi et al., 2020). These methods can all be readily embedded into the proposed ITSRC model as travel time estimators. Based on the work in Kachroo and Sastry (2016), this study adopts a density-based quadratic link travel time estimation model. Accordingly, the travel cost for vehicle k choosing direction d in horizon p is formulated as:

$$T_{i,m,k,d}^p = \begin{cases} \sum_{n=1}^{|Z_{m,k}|} [\xi_1 (\rho_n^p)^2 + \xi_2 \rho_n^p + t_0], & \text{if } Z_{m,k} \neq \emptyset \\ +\infty, & \text{if } Z_{m,k} = \emptyset \end{cases} \quad \forall k \in \mathbf{K}_{i,k}^{\text{ctrl}}, \forall m \in \mathbf{M}_{i,k} \quad (18)$$

where $Z_{m,k}$ denotes the link sequence of the shortest route determined by the A* algorithm. If direction d does not provide a feasible route to the destination, $T_{i,m,k,d}^p$ is assigned an infinite value to indicate infeasibility. ρ_n^p denotes the traffic density on the n -th link during optimization horizon p . The regression is conducted using traffic data generated from microscopic simulations.

4.4. Constraint linearization

As noted in Eqs. (12) and (15), the traffic flow dynamics involve non-linear operators (max and min). To ensure computational tractability, we reformulate these constraints into linear forms.

The queue length update involves a $\max(\cdot)$ operator. Since the objective function minimizes total travel time, we replace the original Eq. (12) with the following linear inequalities:

$$\lambda_{i,m}^p \geq \lambda_{i,m}^{p-1} + q_{i,m}^{\text{in},p} - q_{i,m}^{\text{out},p} \quad (12a)$$

$$\lambda_{i,m}^p \geq 0 \quad (12b)$$

However, the optimal solution $\lambda_{i,m}^p$ still may not be binding on $\lambda_{i,m}^{p-1} + q_{i,m}^{\text{in},p} - q_{i,m}^{\text{out},p}$ or 0. To this end, the $\lambda_{i,m}^p$ should be minimized in the objective function Eq. (5) (Doan & Ukkusuri, 2012). The combination of Eqs. (12a) - (12b) and objective function Eq. (5) is equivalent to original Eq. (12) (Zhu, Chengyuan et al., 2025).

The outflow constraint involves a $\min(\cdot)$ operator. We employ a similar linear formulation:

$$q_{i,m}^{\text{out},p} \leq q_{i,m}^{0,p} + q_{i,m}^{\text{in},p} \quad (15a)$$

$$q_{i,m}^{\text{out},p} \leq S_m (e_{i,m}^p - s_{i,m}^p - t_{\text{loss}}) \quad (15b)$$

Although the linear constraints Eqs. (15a)-(15b) only sets upper bounds, the optimal outflow is guaranteed to be tight. This is driven

by the objective function Eq. (5), which minimizes the queue length $\lambda_{i,m}^p$. As shown in the queue dynamics ($\lambda_{i,m}^p \geq \lambda_{i,m}^{p-1} + q_{i,m}^{\text{in},p} - q_{i,m}^{\text{out},p}$), minimizing $\lambda_{i,m}^p$ is equivalent to maximizing the outflow $q_{i,m}^{\text{out},p}$. Therefore, the model automatically forces $q_{i,m}^{\text{out},p}$ to reach the limit of either demand or capacity, making it equivalent to the original model.

4.5. Model overall formulation

The proposed ITSRC model combines intersection signal control and vehicle routing decisions within an integrated framework. Its objective is to minimize both intersection-level traffic flow delays and individual vehicle routing travel costs. The overall formulation is expressed as Eq. (19):

$$\min J^p = \alpha J_i^p + \beta \sum_{m \in \mathbf{M}_i} \sum_{k \in \mathbf{K}_{i,m}^{\text{ctrl}}} J_k^p \quad (19)$$

Subject to:

Signal plan constraints $\mathbb{C}_S$: Eqs. (7)–(12), Eqs. (14)–(15)

Vehicle routing constraints $\mathbb{C}_R$: Eqs. (16)–(18)

Coupling constraints $\mathbb{C}_C$: Eq. (13)

where α and β are weighting factors to balance the objectives of signal optimization and vehicle routing. In this study, we set $\alpha + \beta = 1$. It is important to note that during the actual optimization process, the non-linear constraints for queue evolution Eq. (12) and outflow transmission Eq. (15) are replaced by their linearized reformulations (Eqs. (12a) - (12b) and Eqs. (15a) - (15b), respectively) to ensure the model constitutes a solvable Mixed-Integer Linear Program.

5. Parallel solution algorithm

In Eq. (19), the objective function is separable across CV and signal controllers, and each controller is associated only with its own control variables and local constraints. This structural property makes the ITSRC problem naturally amenable to decomposition-based solution methods.

Conventional decomposition algorithms, such as Lagrangian relaxation, Benders decomposition, and ADMM, have been widely applied to large-scale coordinated control problems (Niroumand et al., 2025). However, most existing decomposition algorithms adopt a sequential update scheme, in which subproblems are solved one after another, and each controller must wait for the updates of others. To fully exploit the ITSRC decomposition property, a Jacobi-type parallel update scheme is adopted at the subproblem level within an ADMM-based decomposition framework, in which all controllers update their primal variables simultaneously in each iteration via shared consensus variables (Boyd et al., 2011). In our proposed algorithm, each controller optimizes only its own subproblem in parallel and does not require intermediate updates from other controllers (Adeli & Kamal, 1993; Saleh & Adeli, 1994). Algorithm 1 presents the algorithm framework.

5.1. Reformulation of the consensus problem

We use $\mathbb{V}_k$ to denote the CV controller for vehicle k (where $k \in \mathbf{K}_{i,m}$), and $\mathbb{S}_i$ to denote the signal controller for intersection i . Additionally, we define $\mathbb{E}_{i,k}$ as the RSU computation node. To address the coupling between signal timing and route choice, the original ITSRC model is decomposed into sub-problems distributed among these local controllers ($\mathbb{V}_k, \mathbb{S}_i$) and RSUs ($\mathbb{E}_{i,k}$).

According to model formulation (19), the signal controller $\mathbb{S}_i$ is associated with the optimization of the signal plan variable $u_s^{\mathbb{S}_i}$, which consists of the signal timing parameters $s_{i,n}^p$ and $e_{i,n}^p$. The CV controller is associated with the optimization of the vehicle decision variable $u_k^{\mathbb{V}_k}$ for CV $k \in \mathbf{K}_{i,m}^{\text{ctrl}}$, which comprises the travel cost variable $\zeta_{i,m,k}^p$ and the route selection variable $\sigma_{i,m,k,d}$. The RSUs are associated with the optimization of the auxiliary coupling variables $u_s^{\mathbb{E}_{i,k}}$ and $u_k^{\mathbb{E}_{i,k}}$ for all $(i, k) \in \mathcal{E}_i$, where

Algorithm 1 Parallel Algorithm for ITSRG.

Input: Reformulated ITSRG model (23); penalty ρ ; tolerance ϵ ; max iterations $N_{\max}$

Output: Optimal Signal plans $\{u_s^{\mathbb{S}_i}\}$; Optimal CV routes $\{u_k^{\mathbb{V}_k}\}$

```

1: Initialization
2: Initialize iteration counter  $\eta \leftarrow 0$ .
3: Initialize primal variables:  $u_s^{\mathbb{S}_i,0}, u_k^{\mathbb{V}_k,0}$  (Local) and  $u_s^{\mathbb{E},0}, u_k^{\mathbb{E},0}$  (RSU).
4: Initialize consensus variables:  $z_s^0, z_k^0$ .
5: Initialize dual variables:  $\lambda_s^{\mathbb{S}_i,0}, \lambda_k^{\mathbb{V}_k,0}, \lambda_s^{\mathbb{E},0}, \lambda_k^{\mathbb{E},0}$ .

6: while  $\eta < N_{\max}$  do
7:   Step 1: Parallel Primal Updates
8:   Solved simultaneously by all controllers
9:   Signal Controller  $\mathbb{S}_i$ :
10:   Update  $u_s^{\mathbb{S}_i,\eta+1}$  by minimizing local objective via Eq. (25).
11:   CV Controllers  $\mathbb{V}_k$  (for all  $k$ ):
12:   Update  $u_k^{\mathbb{V}_k,\eta+1}$  by minimizing travel cost via Eq. (26).
13:   RSU Nodes:
14:   Update auxiliary variables  $u_s^{\mathbb{E},\eta+1}, u_k^{\mathbb{E},\eta+1}$  via projection
      Eq. (27).
15:   Step 2: Consensus Variable Update
16:   Update global consensus  $z_s^{\eta+1}$  and  $z_k^{\eta+1}$  by averaging local and
      RSU proposals via Eqs. (28)–(29).
17:   Step 3: Dual Variable Update
18:   Update local duals  $\lambda_s^{\mathbb{S}_i,\eta+1}, \lambda_k^{\mathbb{V}_k,\eta+1}$  via Eqs.(30)–(31).
19:   Update RSU duals  $\lambda_s^{\mathbb{E},\eta+1}, \lambda_k^{\mathbb{E},\eta+1}$  via Eqs.(32)–(33).
20:   Step 4: Convergence Check
21:   Calculate primal residual  $r^{\eta+1}$  using max-norm via Eq. (34).
22:   If  $r^{\eta+1} \leq \epsilon$  then
23:     Break
24:   End If
25:    $\eta \leftarrow \eta + 1$ 
26: end while
27: return  $u_s^{\mathbb{S}_i,\eta}$  and  $\{u_k^{\mathbb{V}_k,\eta}\}$ .

```

$\mathcal{E}_i$ denotes the set of intersection i with all controlled vehicles k (i.e., $k \in \mathbf{K}_{i,m}^{\text{ctrl}}$). The auxiliary coupling variables in the RSU node are introduced to enforce the coupling constraints between signal and vehicle routing.

We introduce three indicator functions, $\mathbb{I}_S(\cdot)$, $\mathbb{I}_R(\cdot)$, and $\mathbb{I}_C(\cdot)$, to describe the feasible sets of the signal timing plan ($\mathbb{C}_S$), vehicle route ($\mathbb{C}_R$), and the coupling constraints ($\mathbb{C}_C$), respectively:

$$\mathbb{I}_S(u_s^{\mathbb{S}_i}) = \begin{cases} 0, & \text{if } u_s^{\mathbb{S}_i} \in \mathbb{C}_S \\ +\infty, & \text{otherwise} \end{cases} \quad (20)$$

$$\mathbb{I}_R(u_k^{\mathbb{V}_k}) = \begin{cases} 0, & \text{if } u_k^{\mathbb{V}_k} \in \mathbb{C}_R \\ +\infty, & \text{otherwise} \end{cases} \quad (21)$$

$$\mathbb{I}_C(u_s^{\mathbb{E}_{i,k}}, u_k^{\mathbb{E}_{i,k}}) = \begin{cases} 0, & \text{if } (u_s^{\mathbb{E}_{i,k}}, u_k^{\mathbb{E}_{i,k}}) \in \mathbb{C}_C \\ +\infty, & \text{otherwise} \end{cases} \quad (22)$$

where $\mathbb{C}_S$ and $\mathbb{C}_R$ represent the local constraints for signals and vehicles, respectively. $\mathbb{C}_C$ denotes the set of coupling constraints that are specifically handled by the RSU node $\mathbb{E}_{i,k}$.

By introducing global consensus variables $\mathcal{Z} = \{z_s, z_k\}$, the original problem is reformulated as the following global consensus optimization problem (Boyd et al., 2011):

$$\min_{U, \mathcal{Z}} \quad \hat{J}^p = \underbrace{\left[\alpha J_i^p(u_s^{\mathbb{S}_i}) + \mathbb{I}_S(u_s^{\mathbb{S}_i}) \right]}_{\text{Objective of Signal Controller}}$$

$$+ \underbrace{\sum_{k \in \mathbf{K}_{i,m}} \left[\beta J_k^p(u_k^{\mathbb{V}_k}) + \mathbb{I}_R(u_k^{\mathbb{V}_k}) \right]}_{\text{Objective of CV Controllers}} + \underbrace{\sum_{(i,k) \in \mathcal{E}_i} \mathbb{I}_C(u_s^{\mathbb{E}_{i,k}}, u_k^{\mathbb{E}_{i,k}})}_{\text{Coupling Constraints at RSU Nodes}} \quad (23)$$

$$\text{subject to: } u_s^{\mathbb{S}_i} = z_s, \quad u_s^{\mathbb{E}_{i,k}} = z_s, \quad \forall (i,k) \in \mathcal{E}_i \\ u_k^{\mathbb{V}_k} = z_k, \quad u_k^{\mathbb{E}_{i,k}} = z_k, \quad \forall k \in \mathbf{K}_{i,m}$$

where $u_s^{\mathbb{S}_i}$ and $u_k^{\mathbb{V}_k}$ are the local decision variables optimized by the signal controller and CV k , respectively. $u_s^{\mathbb{E}}$ and $u_k^{\mathbb{E}}$ are the auxiliary variables maintained by the RSU nodes to enforce coupling constraints.

The scaled augmented Lagrangian function of Eq. (23) is given by Eq. (24), where $\lambda = [\lambda^{\mathbb{S}}, \lambda^{\mathbb{V}}, \lambda^{\mathbb{E}}]$ represents the scaled dual variables, and ρ is the penalty parameter. Note that for brevity, we denote the generalized local objectives (including local constraints) as $J^{\mathbb{S}_i}(\cdot) = \alpha J_i^p(\cdot) + \mathbb{I}_S(\cdot)$ and $J^{\mathbb{V}_k}(\cdot) = \beta J_k^p(\cdot) + \mathbb{I}_R(\cdot)$.

$$L_\rho = \underbrace{J^{\mathbb{S}_i}(u_s^{\mathbb{S}_i}) + \frac{\rho}{2} \|u_s^{\mathbb{S}_i} - z_s + \lambda_s^{\mathbb{S}_i}\|_2^2}_{1. \text{ Signal Controller (Local Update)}} + \underbrace{\sum_{k \in \mathbf{K}_{i,m}} \left[J^{\mathbb{V}_k}(u_k^{\mathbb{V}_k}) + \frac{\rho}{2} \|u_k^{\mathbb{V}_k} - z_k + \lambda_k^{\mathbb{V}_k}\|_2^2 \right]}_{2. \text{ CV Controllers (Local Update)}} + \underbrace{\sum_{(i,k) \in \mathcal{E}_i} \left[\mathbb{I}_C(u_s^{\mathbb{E}_{i,k}}, u_k^{\mathbb{E}_{i,k}}) + \frac{\rho}{2} \|u_s^{\mathbb{E}_{i,k}} - z_s + \lambda_s^{\mathbb{E}_{i,k}}\|_2^2 + \frac{\rho}{2} \|u_k^{\mathbb{E}_{i,k}} - z_k + \lambda_k^{\mathbb{E}_{i,k}}\|_2^2 \right]}_{3. \text{ Coupling Constraints (RSU node Update)}} \quad (24)$$

5.2. Parallel distributed algorithm

In the consensus formulation (23), each controller can solve its subproblem without the need for other controllers (Adeli & Kumar, 1995). Therefore, we can use the Jacobi-type parallel update scheme to solve the reformulated consensus problem (Deng et al., 2017). Unlike the Gauss–Seidel update scheme, which commonly requires subproblems to be solved sequentially, the proposed Jacobi-type update scheme enables parallel subproblem updates (Shengbo Eben et al., 2020). Specifically, the Signal Controller, CV Controllers, and RSU Nodes solve their respective subproblems simultaneously at each iteration based on the global consensus variables obtained from the previous iteration. In this manner, the circular dependency between signal timing and traffic flow is handled through consensus coordination rather than sequential coupling. At each iteration η , the algorithm performs the following four steps:

Step 1: Parallel Primal Variable Updates

In this step, the Signal Controller, CV Controllers, and RSU Nodes solve their respective subproblems simultaneously in parallel. For initialization, $u_s^{\mathbb{S}_i,0}$ adopts a fixed-time plan, while $u_k^{\mathbb{V}_k,0}$ uses the current routes. The consensus variables z^0 are initialized as the average of these primal variables, with dual variables λ^0 set to 0 to provide a feasible start.

Signal Controller Update: The signal controller $\mathbb{S}_i$ minimizes its local objective to update the signal plan variable $u_s^{\mathbb{S}_i,\eta+1}$:

$$u_s^{\mathbb{S}_i,\eta+1} = \underset{u_s}{\text{argmin}} \left\{ J^{\mathbb{S}_i}(u_s^{\mathbb{S}_i}) + \frac{\rho}{2} \|u_s^{\mathbb{S}_i} - z_s^\eta + \lambda_s^{\mathbb{S}_i,\eta}\|_2^2 \right\} \quad (25)$$

CV Controller Update: Each CV controller k minimizes its travel cost to update the route variable $\mathbf{u}_k^{\nabla, \eta+1}$:

$$\mathbf{u}_k^{\nabla, \eta+1} = \underset{\mathbf{u}_k}{\operatorname{argmin}} \left\{ J^{\nabla}_k(\mathbf{u}_k^{\nabla}) + \frac{\rho}{2} \left\| \mathbf{u}_k^{\nabla} - \mathbf{z}_k^{\nabla} + \lambda_k^{\nabla, \eta} \right\|_2^2 \right\}, \quad \forall k \in K_{i,m} \quad (26)$$

RSU Node Update: The RSU solves a projection problem to satisfy the coupling constraints $\mathbb{C}_C$. It updates the auxiliary variables $\mathbf{u}_s^{\mathbb{E}}$ and $\mathbf{u}_k^{\mathbb{E}}$:

$$\left\{ \mathbf{u}_s^{\mathbb{E}, \eta+1}, \mathbf{u}_k^{\mathbb{E}, \eta+1} \right\} = \underset{\mathbf{u}^{\mathbb{E}}}{\operatorname{argmin}} \left\{ \mathbb{I}_C(\mathbf{u}_s^{\mathbb{E}}, \mathbf{u}_k^{\mathbb{E}}) + \frac{\rho}{2} \left\| \mathbf{u}_s^{\mathbb{E}} - \mathbf{z}_s^{\mathbb{E}} + \lambda_s^{\mathbb{E}, \eta} \right\|_2^2 + \sum_{k \in K_{i,m}} \frac{\rho}{2} \left\| \mathbf{u}_k^{\mathbb{E}} - \mathbf{z}_k^{\mathbb{E}} + \lambda_k^{\mathbb{E}, \eta} \right\|_2^2 \right\} \quad (27)$$

Note that Eq. (25) is a standard convex quadratic programming problem, which could be solved with existing optimization algorithms, e.g., the interior point method or dual methods (Zhang et al., 2025). However, Eq. (26) and Eq. (27) serve as non-convex optimization problems with binary variables. Since these subproblems only involve decision variables of CVs, their solution space remains limited. Consequently, they can be efficiently solved using standard branch-and-bound-based solvers (Gau & Schrage, 2004; Niebling & Eichfelder, 2019). Existing studies have also widely applied ADMM to similar non-convex problems involving discrete variables, demonstrating good practical performance and convergence properties under certain conditions (Wang et al., 2019).

Step 2: Consensus Variable Update

The global consensus variables $\mathbf{z}^{\eta+1}$ are updated by balancing the proposals from the local controllers and the RSU feasible solutions:

$$\mathbf{z}_s^{\eta+1} = \frac{1}{1+1} \left[\left(\mathbf{u}_s^{\nabla, \eta+1} + \lambda_s^{\nabla, \eta} \right) + \left(\mathbf{u}_s^{\mathbb{E}, \eta+1} + \lambda_s^{\mathbb{E}, \eta} \right) \right] \quad (28)$$

$$\mathbf{z}_k^{\eta+1} = \frac{1}{1+1} \left[\left(\mathbf{u}_k^{\nabla, \eta+1} + \lambda_k^{\nabla, \eta} \right) + \left(\mathbf{u}_k^{\mathbb{E}, \eta+1} + \lambda_k^{\mathbb{E}, \eta} \right) \right], \quad \forall k \in K_{i,m} \quad (29)$$

In the CV controller, the consensus variable $\mathbf{z}_k$ corresponds to the travel cost $\zeta_{i,m,k}^p$. This choice is made because $\zeta_{i,m,k}^p$ is continuous, which facilitates the convergence of the ADMM consensus updates (Wang et al., 2019). Since Eq. (13) establishes a one-to-one correspondence between $\zeta_{i,m,k}^p$ and $\sigma_{i,m,k,d}^p$, the consensus variable update enforces consistency in travel cost variable $\zeta_{i,m,k}^p$ and CV route selection variable $\sigma_{i,m,k,d}^p$.

Step 3: Dual Variable Update

The scaled dual variables are updated in parallel:

$$\lambda_s^{\nabla, \eta+1} = \lambda_s^{\nabla, \eta} + \left(\mathbf{u}_s^{\nabla, \eta+1} - \mathbf{z}_s^{\nabla, \eta+1} \right) \quad (30)$$

$$\lambda_k^{\nabla, \eta+1} = \lambda_k^{\nabla, \eta} + \left(\mathbf{u}_k^{\nabla, \eta+1} - \mathbf{z}_k^{\nabla, \eta+1} \right) \quad (31)$$

$$\lambda_s^{\mathbb{E}, \eta+1} = \lambda_s^{\mathbb{E}, \eta} + \left(\mathbf{u}_s^{\mathbb{E}, \eta+1} - \mathbf{z}_s^{\mathbb{E}, \eta+1} \right) \quad (32)$$

$$\lambda_k^{\mathbb{E}, \eta+1} = \lambda_k^{\mathbb{E}, \eta} + \left(\mathbf{u}_k^{\mathbb{E}, \eta+1} - \mathbf{z}_k^{\mathbb{E}, \eta+1} \right) \quad (33)$$

Step 4: Stopping Criterion

The algorithm iterates until the primal residuals satisfy the tolerance ϵ :

$$r^{\eta+1} = \max \left\{ \left\| \mathbf{u}_s^{\nabla, \eta+1} - \mathbf{z}_s^{\nabla, \eta+1} \right\|_2, \max_{k \in K_{i,m}} \left\| \mathbf{u}_k^{\nabla, \eta+1} - \mathbf{z}_k^{\nabla, \eta+1} \right\|_2 \right\} \leq \epsilon \quad (34)$$

The criterion employs the maximum norm to enforce strict consensus across all controllers, avoiding constraint violations from local outliers. The tolerance is set to $\epsilon = 1$ to balance computational efficiency with practical engineering granularity, as sub-second precision yields negligible operational benefits in traffic control contexts.

The proposed ITSRG requires iterative data exchange between vehicles and the RSU within each control cycle. The communication load still remains physically feasible under existing V2X standards. Each vehicle transmits a lightweight packet comprising its ID, position, velocity, destination, and route, and the RSU returns a signal plan of comparable size. Under DSRC (IEEE 802.11p) or C-V2X standards, which support channel capacities from several Mbps to hundreds of Mbps, the total data volume is only a small fraction of the available channel bandwidth (Lim et al., 2021). Therefore, the ITSRG is communicatively viable within the existing deployed V2X infrastructure.

6. Evaluation

In this section, we present the evaluation results obtained from a simulation based on a real-world road network for the proposed ITSRG model and benchmarks. Moreover, the sensitivity analyses are performed to further validate the effectiveness of the ITSRG model in different traffic situations.

6.1. Experiment design

This study evaluated the ITSRG in the real-world road network of Qinzhou City, China, in the SUMO traffic simulation platform. As shown in Fig. 4, this road network includes 25 signalized intersections, both cross-shaped and T-shaped. The duration of each simulation session is set to 1800 s, with the initial 300 s serving as a warm-up period. Following the research in (Chunhui et al., 2018; Zhu et al., 2025), Table 2 summarizes the main parameters employed in the ITSRG. Unless routing optimization is included in the control strategy, vehicles adhere to their pre-assigned routes throughout the simulation.

The traffic demand was collected in the morning peak hour on 20 December 2023, yielding a total volume of 5400 vehicles per hour on the road network. The demand profile corresponds to a high-demand scenario and is used to evaluate ITSRG effectiveness. In the sensitivity analysis section, this study performs the ITSRG under five different demand levels: low, medium, high, saturated, and over-saturated, which will be detailed in Section 6.4.1. In the evaluation, this study assumes that all vehicles are CVs and will obey the optimized route decisions.

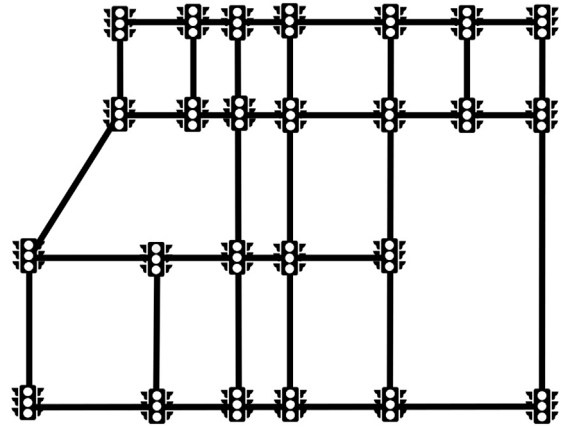


Fig. 4. The real-world road network of Qinzhou.

Table 2

Parameter setting.

Parameter	Value	Parameter	Value
The minimum green time g_{min}	10s	Weighting factor α	0.9
The maximum green time g_{max}	40s	Weighting factor β	0.1
Rolling time window length	25s	Penalty parameter ρ	100
Inter-green time ψ	3s	Tolerance of stopping criterion ϵ	1

This study will further evaluate the model's performance under different compliance rates in Section 6.4.2.

6.2. Benchmark implementation

In our experiment, the performance and effectiveness of the ITSRG are evaluated with several benchmarks. All optimization problems involved in these benchmarks are solved using the Gurobi solver:

- **FSR (Fixed Signal and Route).** This benchmark represents the current real-world operation without optimization. Both signal timings and vehicle routes remain fixed throughout the simulation and are not updated in response to real-time traffic conditions.
- **DUE (Dynamic User Equilibrium).** This benchmark adopts a traditional dynamic routing strategy based on the dynamic user equilibrium principle (Dial, 2006). At each decision step, vehicle route choices are updated according to real-time link travel times and current network conditions. The signal timings remain fixed and identical to those in FSR.
- **BP (Back-Pressure signal optimization method):** This benchmark applies a back-pressure-based signal control algorithm (Gao & Zhang, 2022). Signal phase durations are updated at each decision step based on observed queue lengths and turning flows. The vehicle routes remain fixed and identical to those in the FSR.
- **DUE-BP:** This benchmark represents a sequential coordination framework commonly used in the literature (Chen, Fan et al., 2022). In the first stage, vehicle routes are dynamically updated based on the DUE method. In the second stage, the resulting turning rates are passed to a BP signal controller, which updates signal timings independently. The comparison between ITSRG and DUE-BP evaluates the benefits of integrated formulation.
- **SRAO (Signal-Route Alternating Optimization):** This benchmark adopts a two-stage iterative process following the classic alternating optimization framework for signal-route coordination (Chen & Qiu, 2022; Liu et al., 2025). In the first stage, a dual-ring phase model is employed to update signal timings, which then informs the dynamic route updates in the second stage based on the SO. This alternating sequence between signal and route optimization continues until convergence is reached. This benchmark treats signal control and route guidance as independent processes, and consequently, it does not constitute an integrated optimization framework.
- **ITSRG:** The proposed integrated optimization model in this study. The ITSRG is solved by the parallel solution algorithm for real-time optimization.

6.3. Results

6.3.1. Performance of traffic management

Fig. 5 illustrates the distribution of vehicle delay under different traffic management strategies in the real-world network. Delay is measured relative to free-flow travel time, allowing a fair comparison. The traffic management method that only considers route planning (i.e., DUE) or signal control (i.e., BP) can improve traffic efficiency compared with the FSR baseline. The proposed ITSRG achieved the shortest average delay among all benchmarks. The proposed ITSRG reduced average delay by 22.7% compared with SRAO. This result highlights the benefit of simultaneously optimizing traffic signals and route guidance.

As shown in Fig. 5, the delays of six traffic control strategies are compared. The entire distribution curve of the proposed ITSRG shifts left relative to all benchmark strategies, indicating a significant reduction in delay. In addition, ITSRG shows fewer long-tail distributions and anomalous peaks, indicating that it effectively avoids extreme congestion for individual vehicles. These results further confirm that the ITSRG yields more stable and efficient network performance.

Fig. 6 shows the spatial distribution of link queues at several representative time steps during the simulation. The color scale represents the

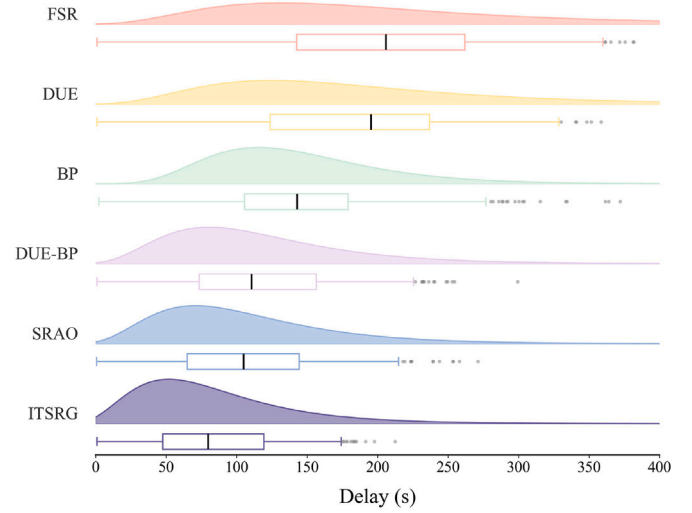


Fig. 5. Vehicle delay distributions for different benchmarks in the high-demand scenario.

queue length on each link. Compared with benchmark strategies, ITSRG produces fewer links with long queues and shorter durations of high queue levels. In particular, ITSRG avoids the emergence of congestion that persists under other benchmark strategies. This improvement results from a more balanced spatial distribution of the queue under ITSRG, which prevents traffic demand from concentrating on a few critical links. The comparison demonstrates that integrating signal control and vehicle routing yields significant benefits for mitigating network-level congestion.

Table 3 compares the rerouting behavior of DUE, DUE-BP, SRAO, and ITSRG, using FSR as the baseline. The analysis focuses exclusively on vehicles that changed their routes, which are called rerouted vehicles. Under the DUE, the average travel time of rerouted vehicles increases by 15.63s. DUE initially guides many vehicles toward less congested routes; however, these routes soon become congested again. Although this approach can temporarily relieve congestion in certain parts of the network, the rerouted vehicles may experience longer travel times as congestion redistributes across alternative routes. The DUE-BP method performs slightly better than DUE; however, its travel time savings are much smaller than those of ITSRG. This indicates that although DUE-BP can consider the signal impact on vehicle routing compared to DUE, its separated optimization structure still cannot ensure effective coordination between signal control and vehicle routing. The SRAO further improves the alignment between arrival flows and signal timing through iterative updates. However, its sequential framework remains vulnerable to local optima, which restricts overall performance gains.

The proposed ITSRG model achieves a rerouting frequency of 1.47 and a probability of 53.12% per complete trip, indicating that vehicles dynamically adjust their routes in response to real-time traffic conditions. More importantly, ITSRG achieves the greatest travel time reduction among rerouted vehicles (−101.72s), demonstrating that even though some vehicles travel longer distances, their total travel time is reduced due to smoother traffic conditions. This confirms the effectiveness of simultaneously optimizing signal timing and vehicle route guidance in mitigating congestion and improving overall network efficiency.

6.3.2. Computational efficiency

In this section, this study evaluates the computational scalability of the proposed parallel distributed solution algorithm. This study uses the centralized solution as the benchmark, which directly solves the centralized problem (19).

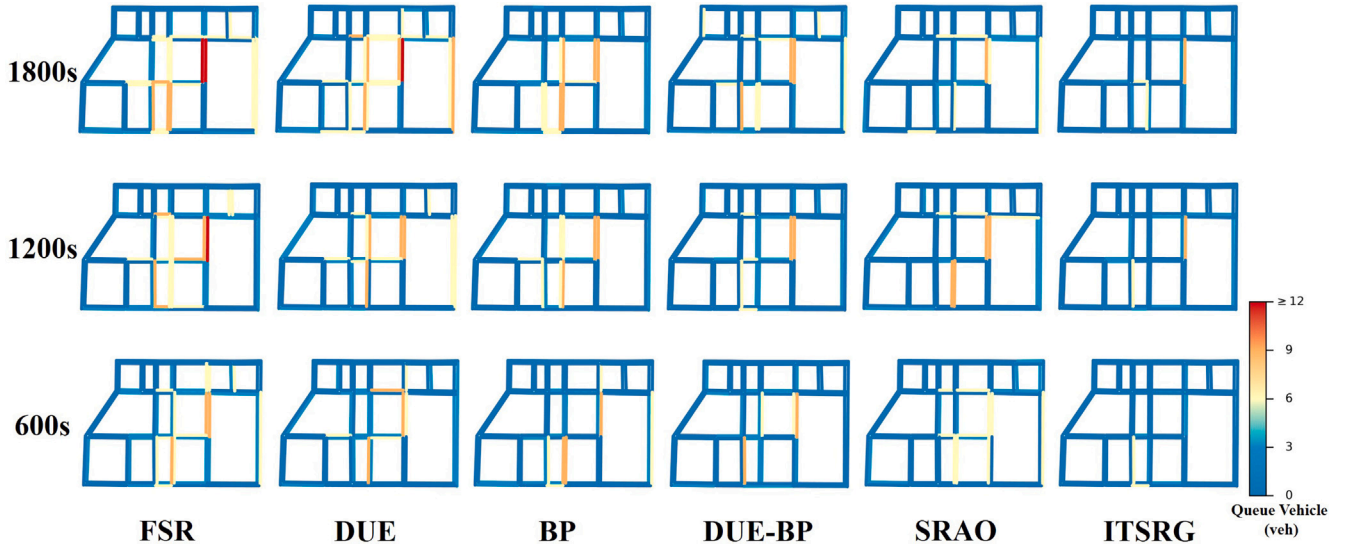


Fig. 6. Network queue distribution for different benchmarks in the high-demand scenario.

Table 3

Comparison of rerouting behavior in DUE, DUE-BP, and ITSRG.

Method	Rerouted frequency	Rerouted probability	Vehicle travel time saving vs. FSR /s
DUE	0.70	50.30%	+15.63
DUE-BP	1.56	60.10%	-33.71
SRAO	1.13	57.85%	-39.46
ITSRG	1.47	53.12%	-101.72

Table 4

Comparison of algorithm solving performance.

Solution Type	Controller	Average computation time/s	Maximum computation time/s
Parallel Distributed Solution	Vehicle	0.036	0.041
	Signal	0.093	0.207
Centralized	–	1.878	3.070

Table 4 shows the average and maximum computation times of these two solution methods. In the parallel distributed solution algorithm, the average computation time is 0.036s for CV controllers and 0.093s for signal controllers, with maximum values of 0.041s and 0.207s, respectively. In contrast, the centralized solution needs 1.878s on average, with the maximum time of 3.070s. These results demonstrate the significant computational advantage of the proposed parallel distributed solution. By decomposing the integrated optimization problem and distributing the computation across CV and signal controllers, the algorithm achieves computation times under 0.3s, enabling real-time implementation. By decomposing the integrated optimization problem into intersection-level subproblems and allocating computation to CV and signal controllers, the algorithm limits the size of each local optimization task. Because each intersection is solved independently in parallel, computation time scales with local complexity (analyzed in Section 6.4.1) instead of network size, ensuring real-time performance even as the network grows.

Fig. 7 further illustrates the convergence performance of the parallel distributed algorithm. It displays the primal residuals of the control variables for one CV controller during a single example optimization in the high-demand condition. The decreasing trend in primal residuals

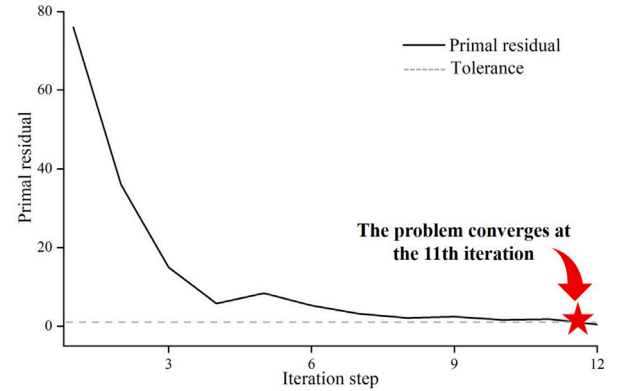


Fig. 7. The iterations process of the proposed parallel distributed algorithm in an example scenario.

indicates that the local and consensus variables are progressively aligning. As the number of iterations increases, the residuals progressively decrease and eventually stabilize, indicating convergence toward the optimal solution. In this case, convergence is achieved at the 11th iteration. Across our experiments, the optimization typically converges within approximately 10 iterations, demonstrating the algorithm's efficiency and robustness in real-world applications.

6.4. Sensitivity analysis

6.4.1. Traffic demand level

This study evaluates the efficiency of ITSRG and benchmarks under multiple traffic demand levels. Five demand levels are designed, ranging from low to over-saturated conditions. The total numbers of trips are 4250, 4800, 5500, 6250, and 7250 vehicles per hour for the low, medium, high, saturated, and over-saturated demand levels, respectively. Fig. 8 compares the average travel times of the benchmark methods and ITSRG across the five demand levels. As traffic demand increases, the average travel times of all methods rise, while ITSRG consistently maintains lower values than the others. Under over-saturated conditions, ITSRG achieves an average travel time of 277.4s, compared to 429.8s, 399.1s, 376.9s, 317.8s, and 305.2s for FSR, DUE, BP, DUE-BP, and SRAO, respectively. Even under over-saturated conditions, ITSRG effectively mitigates congestion propagation and keeps the increase

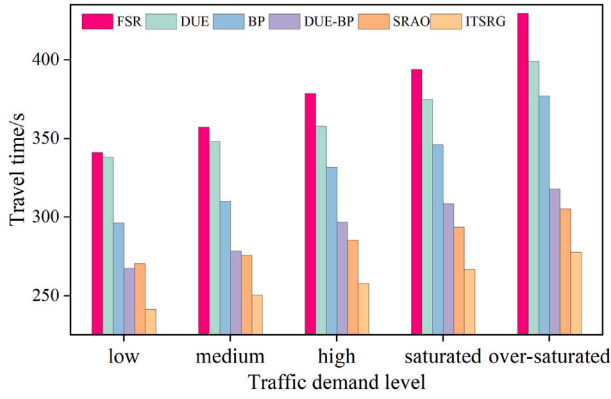


Fig. 8. Comparison of travel time under various traffic demand levels.

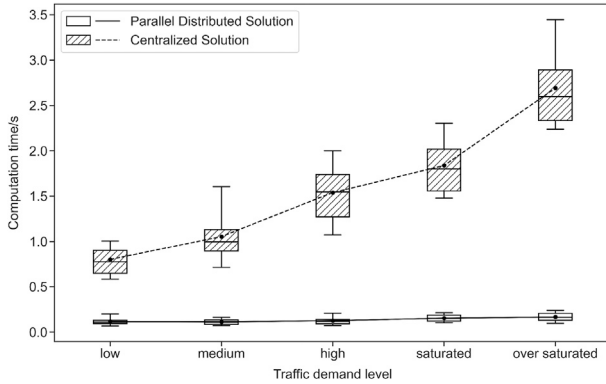


Fig. 9. Total computation time under various traffic demand levels.

in travel time minimal. These results demonstrate the robustness and superiority of ITSRG across various traffic conditions.

We further evaluate the computation time of the proposed parallel distributed algorithm under different demand levels. Fig. 9 illustrates the total computation time required for each iteration process. As shown in the figure, the optimization time of the proposed parallel distributed algorithm remains nearly constant even as traffic demand increases. Across all five demand levels, the average computation time stays around 0.2s, with only minor variations indicated by the box plots. However, directly solving the centralized model (19) shows a pronounced exponential-like growth in computation time. The computation time of the centralized solution grows from approximately 0.8s under low demand to more than 2.0s under saturated demand, and even exceeds 3.0s in the over-saturated scenario. This indicates the limited scalability of the centralized approach as traffic conditions become increasingly congested. These results confirm that the proposed parallel distributed framework offers significantly superior computational efficiency and scalability. It is well-suited for real-time applications in large-scale scenarios.

6.4.2. Route guidance compliance rate

The route guidance compliance rate represents the probability that a vehicle executes a rerouting instruction issued by the control system. In this study, the compliance rate is introduced to capture execution uncertainty at the control level, which may arise from imperfect communication, delayed responses, or drivers' refusal to follow route guidance. Sensitive experiments are evaluated in various compliance rates from 30% to 100% across five distinct traffic demand levels.

Fig. 10 plots the compliance rate against the average travel time, revealing that the effectiveness of the ITSRG is highly correlated with the

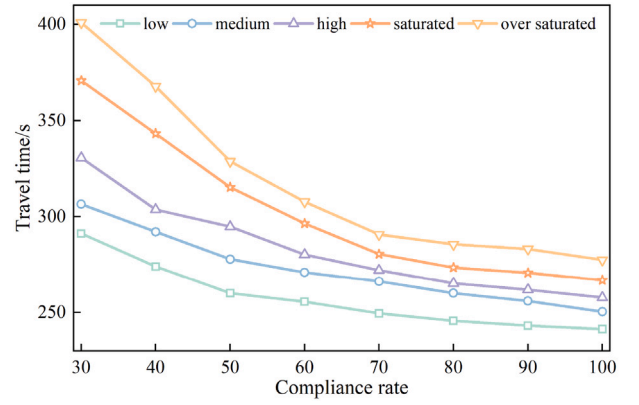


Fig. 10. Impacts of vehicle routing compliance rates.

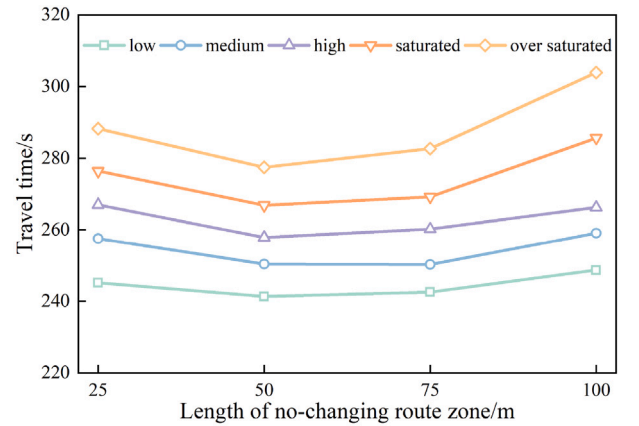


Fig. 11. Vehicle travel time under various lengths of no-changing-route zones.

compliance rate. Across all demand levels, an increase in compliance rate leads to a decrease in travel time. Under the low demand level, increasing the compliance rate from 30% to 100% yields a substantial decrease in average travel time, representing a 17.13% improvement. The benefits of higher compliance are further amplified as traffic demand escalates. For the over-saturated scenario, the average travel time decreases from 30% compliance rate to full compliance, a remarkable reduction of 30.79%, highlighting the critical role of vehicle rerouting in enhancing network efficiency under severe congestion scenarios.

Moreover, the non-linear relationship between compliance rate and travel time reduction is shown in Fig. 10. The most substantial gains in performance are achieved in the lower compliance rates (i.e., from 30% to 60%). For instance, under the over-saturated demand level, increasing compliance from 30% to 60% lowers the average travel time by 93.13s. In contrast, the increase from 70% to 100% in the compliance rate only contributes to a smaller reduction in travel time. This implies that significant improvements can be achieved even when only a portion of drivers comply, indicating the practical ability of the proposed ITSRG in the real world.

6.4.3. The length of no-changing-route zone

To operationalize the concept of the no-changing-route zone and examine its impact on system performance, we conduct a sensitivity analysis with respect to its length, denoted by L^{NC} . We evaluate four candidate values of $L^{NC} \in (25, 50, 75, 100)$ m under five demand levels (low, medium, high, saturated, and over-saturated).

As shown in Fig. 11, the system performance exhibits a clear convex trend with respect to the length of the no-changing-route zone. When

L^{NC} is too short (e.g., 25 m), the average delay is consistently higher than that under the baseline setting of 50 m across all demand levels. Although a shorter zone enlarges the theoretical routing optimization space, it neglects lane-changing kinematic constraints. Vehicles reassigned to new lanes often lack sufficient distance to safely complete the maneuver, leading to forced merges, local turbulence near the stop line, and occasional infeasible routing outcomes, which collectively degrade operational efficiency. Conversely, when L^{NC} becomes excessively long (e.g., 100 m), the average delay increases markedly, particularly under saturated and over-saturated conditions (e.g., from 280.45 s to 301.88 s in the over-saturated scenario). In this case, vehicles are required to commit to their turning movements too early, substantially reducing the size of the controllable vehicle set $K_{i,m}^{ctrl}$. This premature locking limits the controller's ability to adapt routing decisions to evolving traffic conditions and demand fluctuations.

Overall, the results indicate a clear trade-off between routing flexibility and execution feasibility. A moderate, no-changing-route zone length (50–75 m) strikes a balance between these two effects and yields the best overall performance across traffic regimes. While the sensitivity analysis presented here provides theoretical insights into the effect of L^{NC} on model performance, the appropriate value in practice should be calibrated according to site-specific conditions, including operating speed, demand level, and geometric characteristics of the roadway (Xue et al., 2025).

6.4.4. Weighting coefficient

The proposed ITSRG aims to balance individual vehicle travel cost with intersection traffic conditions. Within the model, this balance is regulated by the weighting coefficient β in the objective function in Eq. (19). A larger β emphasizes minimizing individual travel costs and promotes a conservative routing strategy that keeps vehicles on shorter routes. Conversely, a smaller β focuses on reducing queue lengths and delays at the intersection, which may guide vehicles onto longer routes to bypass local congestion.

To evaluate the influence of β , a sensitivity analysis was conducted under multiple demand levels, as shown in Fig. 12. The results reveal a clear convex relationship between β and the travel time across all traffic demand levels. When β is too small (e.g., 0.01), the model excessively penalizes queuing, leading to an overly aggressive routing policy that increases vehicle travel time through unnecessary detours. In contrast, when β is too large (e.g., 0.2), the model becomes overly conservative and prioritizes shortest paths over network performance. This reluctance to reroute prevents the system from using network capacity efficiently, which increases congestion and travel time.

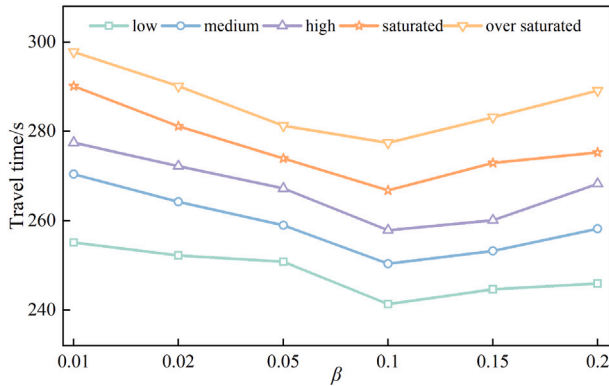


Fig. 12. Impact of routing coefficient.

The analysis shows that an optimal balance is achieved when $\beta = 0.1$. At this value, the algorithm achieves the best network control performance, yielding the lowest average travel times from low demand (241.34s) to over-saturated conditions (277.45s). It effectively balances efficient routing with strategic detours to maintain smooth traffic flow, validating the ITSRG's ability to achieve near-optimal network performance.

7. Conclusion

This study proposes a novel integrated optimization model for coordinating traffic signal control and vehicle routing. The ITSRG model facilitates the interaction between individual vehicle routing decisions and traffic flow dynamics, enabling proactive coordination between routing and signal control. The flexible phase structure further enlarges the feasible solution space of signal optimization, allowing traffic redistribution induced by route guidance to be effectively accommodated. Moreover, the ADMM-based distributed solution functions not merely as a computational tool, but as a coordination mechanism that decouples vehicle-level and signal-level decisions across different time scales, ensuring stable and executable coordination under frequent traffic fluctuations.

Evaluation in a real-world urban road network demonstrates the effectiveness of the ITSRG. The ITSRG consistently outperformed benchmark methods. By jointly optimizing vehicle routing and signal plans, ITSRG accounts for signal delays at intersections and achieves shorter overall travel times, even though some vehicles take slightly longer routes to alleviate congestion. The parallel distributed algorithm exhibits promising computational efficiency, with each subproblem solved in roughly 0.1s. Sensitivity analysis further validates the effectiveness of ITSRG under various traffic scenarios.

Several promising directions could further enhance this study. First, incorporating driver behavioral responses to route guidance would further improve the effectiveness of ITSRG. Since some drivers may be reluctant to follow recommended detours, especially when they appear longer in distance, integrating compliance models or bounded-rational decision mechanisms could help capture heterogeneous acceptance levels and their impact on network performance. Second, extending the ITSRG to multi-modal transportation systems would allow differentiated control strategies for various traffic participants. By introducing priority and coordination mechanisms for public transport, freight, and emergency vehicles, the model could better balance efficiency, equity, and safety in complex urban environments. Finally, in practical applications, communication latency inevitably affects the real-time performance of the proposed algorithm, and future work will be dedicated to explicitly investigating its impact. In addition, extending the proposed framework to mixed traffic environments with heterogeneous vehicle connectivity and compliance behaviors represents an important research direction, thereby improving the practical applicability of the proposed method.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Notations

Table A.1 lists the main notations used throughout this paper.

Table A.1

Main notations.

Sets	
$K_{i,m}$	Set of all vehicles in the movement m at intersection i
$K_{i,m}^{ctrl}$	Set of all vehicles that are in the route optimization zone in the movement m at intersection i
M_i	Set of movements of intersection i
U_m	Set of upstream movements of movement m
$D_{i,k}$	Set of direction of vehicle k in intersection i
Indices	
i	Index of intersection
m, n	Index of movement
k	Index of vehicle
d	Index of direction
Parameters	
$Q_{i,m}^{0,p}$	The initial queue length for movement m at intersection i during the p th optimization horizon
t_k^p	A constant determined by the vehicle's arrival profile.
p_k	the queue position p_k of each queued vehicle k .
g_{min}, g_{max}	The lower and upper bounds of the green signal interval
t_p	The signal control plan is re-optimized periodically at each decision point p
t_{loss}	Time lost due to vehicle start-up.
ψ	The safety time gap between conflicting green durations
S_m	The saturation flow rate
$T_{i,m,k,d}^p$	The shortest route cost to the destination of vehicle k when choosing direction d at movement m in intersection i
ρ_n^p	The traffic density on the n -th link during optimization horizon p .
ξ_1, ξ_2, t_0	The coefficients in the density-based quadratic link travel time estimation model, which are obtained from simulation results regression.
$\gamma_{m,n}$	Binary parameter representing the incompatibility between movements m and n . If movements m and n are incompatible, then $\gamma_{m,n} = 1$; otherwise, $\gamma_{m,n} = 0$
Variables	
τ_k	The vehicle k actual departure time
d_k^p	The delay of vehicle k is the gap between its actual departure and virtual arrival times
$s_{i,m}^p, e_{i,m}^p$	The green start and end times for movement m at intersection i during optimization horizon p .
$\phi_{m,n}$	The binary variable indicating the sequence of movements. If the phase of movement n occurs after movement m , then $\phi_{m,n} = 1$; otherwise, $\phi_{m,n} = 0$.
$\sigma_{i,m,k,d}^p$	The binary variable to represent vehicle k on movement m at intersection i whether choose the direction $d \in D$ in the p th optimization horizon. When the vehicle k selects direction d , then $\sigma_{i,m,k,d}^p = 1$; otherwise, $\sigma_{i,m,k,d}^p = 0$.
$c_{i,m,k}^p$	The travel cost of vehicle k on movement m at intersection i
$q_{i,m}^{in,p}$	The total count of vehicles reaching the stop-line for movement m at intersection i during optimization horizon p .
$q_{i,m}^{out,p}$	The total count of vehicles departing the stop-line for movement m at intersection i during the optimization horizon p .
$\lambda_{i,m}^p$	The residual queue vehicles of movement m carried over to the next optimization horizon

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